



How does passing the 1.5°C threshold create new diplomatic and climate challenges for India?

Limiting Overshoot report • Global Methane Pledge • Bhotekoshi flood disaster • Enteric fermentation • Overshoot scenario

GS Paper III
Environment
Pollution and
Degradation

KNOWLEDGE POINTS

- **Irreversible Overshoot Damage:** Breaching 1.5°C triggers permanent environmental loss that basic adaptation cannot repair. Cooling the planet after crossing safe limits is far harder than stopping emissions now, forcing every country to rethink its core long-term climate survival plans.
- **Growing Methane Scrutiny:** International bodies increasingly pressure nations to slash methane emissions quickly. India has avoided global methane treaties because farming and livestock sustain millions of rural livelihoods, creating intense friction with wealthy countries that focus narrowly on immediate gas cuts.
- **Shifting Diplomatic Perceptions:** Vulnerable neighbors now group India with historical heavy polluters due to its massive total emissions. This complicates India's traditional stance as a climate victim, demanding smarter diplomatic communication to balance rapid domestic development with regional ecological duties.
- **Climate Finance Inequity:** Rich countries historically released most atmospheric carbon but fail to provide promised funding and green technology. Developing nations argue that without adequate adaptation money, demanding rapid emission cuts unfairly threatens economic growth and poverty eradication.
- **Short-Lived Climate Pollutants:** Methane traps far more heat than carbon dioxide over shorter periods, causing one-third of current warming. Slashing short-lived climate pollutants provides immediate cooling benefits, making it an attractive fast-track target for global mitigation treaties.

QUICK FACTS

- UNEP's **Limiting Overshoot report** warns that existing global policies lead to 2.6°C of heating, while fully meeting all national climate pledges still results in 1.8°C of warming.
- Methane contributes approximately 0.5°C of current global warming, prompting scientists to target it as the fastest and most effective way to slow near-term heating compared to carbon dioxide.
- More than 155 countries joined the **Global Methane Pledge** targeting a 30% cut in anthropogenic methane below 2020 levels by 2030, though India has remained outside this agreement.
- Following the catastrophic **Bhotekoshi flood disaster**, Nepal officially urged major industrial emitters including China, the United States, and India to recognize climate damage inflicted on vulnerable Himalayan nations.

KEY TAKEAWAY

As temporary temperature overshoot becomes reality, India must defend its developmental space while proactively addressing short-lived pollutants and reassuring vulnerable neighbors through balanced, transparent, and collaborative climate diplomacy.

MAINS PERSPECTIVE

Evaluate the policy trade-offs India faces between mitigating agricultural methane and safeguarding rural food security and livelihoods?

India's resistance to binding international methane agreements stems from deep structural socio-economic realities, where non-carbon greenhouse gas mitigation directly collides with agricultural resilience, smallholder survival, and national food sovereignty.

Unlike industrialized nations where methane predominantly originates from fossil fuel leaks and commercial landfills, over seventy percent of India's agricultural emissions arise from **enteric fermentation** in livestock and anaerobic decomposition in flooded paddy fields. Imposing rigid, top-down emission reductions risks undermining the agrarian livelihoods of over one hundred million small and marginal farmers who rely on dairy cattle for daily subsistence. Furthermore, aggressively altering wet-paddy irrigation regimes without accessible technology threatens staple grain productivity, creating acute food security vulnerabilities for economically disadvantaged populations. However, total inaction creates severe long-term environmental hazards. Unchecked methane exacerbates tropospheric ozone formation, which directly damages crop yields, while accelerating warming intensifies unpredictable monsoon patterns across the subcontinent. Reconciling this acute tension requires non-coercive techno-economic transitions rather than punitive mandates. Deploying subsidized cattle feed supplements to suppress enteric emissions, scaling direct-seeded rice cultivation, and promoting community-scale biogas digesters allow India to curb emissions while augmenting farm revenues and stabilizing essential staple output.

Navigating this policy trade-off demands prioritizing equitable climate justice over blanket mitigation, ensuring that agricultural modernization enhances rural prosperity and yields while voluntarily lowering emission intensity across vulnerable farming ecosystems.

DAILY MCQs

- With reference to the **Global Methane Pledge**, consider the following statements:
 - It aims to reduce global anthropogenic methane emissions by at least 30 percent below 2020 levels by 2030.
 - India is a founding signatory to this pledge under the United Nations Framework Convention on Climate Change (UNFCCC).
 Which of the statements given above is/are correct?
 - 1 only
 - 2 only
 - Both 1 and 2
 - Neither 1 nor 2
- Consider the following statements
 - regarding methane as a greenhouse gas:
 - Methane is considered a short-lived climate pollutant with a significantly shorter atmospheric lifespan than carbon dioxide.
 - According to international environmental assessments, methane is responsible for approximately 0.5°C of current global warming.
 Which of the statements given above is/are correct?
 - 1 only
 - 2 only
 - Both 1 and 2
 - Neither 1 nor 2
- According to the United Nations Environment Programme's assessment on climate pathways, which of the following best describes an 'overshoot' scenario?
 - Completely abandoning renewable energy transitions in favor of geoengineering solutions.
 - Temporarily exceeding a target temperature threshold like 1.5°C before bringing temperatures back down through long-term removals.
 - Transferring historical emissions debt from developed countries entirely to emerging market economies.
 - Permanently setting global warming targets to 2.6°C to prevent industrial disruption.

Answer Key • 1-A • 2-C • 3-B



Why must public healthcare expand beyond reproductive years to support women facing menopause?

Primary health centers • Workforce participation • Ruthu Thare • geriatric care • Article 21

GS Paper II

Governance

Issues Relating to Health

KNOWLEDGE POINTS

- **Ending Silent Suffering:** Most public healthcare programs only focus on pregnancy and childbirth. Consequently, millions of older women are forced to silently endure severe physical fatigue, emotional distress, and joint pain during menopause without any basic medical support.
- **Routine Healthcare Integration:** Bringing menopause care into routine government health centers ensures early diagnosis. Through regular screening, counseling, and referrals at **primary health centers**, women can access timely treatment for severe symptoms instead of being dismissed or ignored.
- **Sensitizing Medical Staff:** Doctors and families often dismiss menopausal symptoms as normal aging rather than real medical issues. Public health policies help train medical professionals to validate women's complaints, eliminate clinical bias, and provide proper treatment with dignity.
- **Understanding Biological Changes:** Menopause marks the permanent end of menstrual cycles due to falling estrogen and progesterone levels. The transitional perimenopause phase brings hot flashes, mood swings, and sleep disruption that significantly harm daily living.
- **Promoting Gender Equity:** Neglecting non-reproductive health reduces older women's **workforce participation** and quality of life. Expanding public care recognizes health as a lifelong fundamental right, fulfilling constitutional promises of equality and social justice for aging women.

QUICK FACTS

- Menopause is clinically established after twelve consecutive months without menstrual cycles, triggered by ovaries reducing the natural synthesis of essential hormones namely oestrogen and progesterone.
- Karnataka launched **Ruthu Thare** on August 4, establishing the country's first dedicated public health framework addressing mid-life medical challenges like perimenopause and menopause.
- Tamil Nadu announced its comprehensive perimenopausal health policy on August 17 to facilitate early symptom detection, professional counselling, and institutional treatment through primary health centres.
- Both southern state policies institute structured tiered care, connecting grassroots community screening and dedicated well-women clinics to specialized treatments at district hospitals and government medical colleges.

KEY TAKEAWAY

Integrating perimenopause care into routine public healthcare transforms mid-life women's well-being by providing community-level diagnosis, clinical validation, and targeted medical therapy for disruptive physical and psychological transitions.

MAINS PERSPECTIVE

Examine how shifting from reproductive-centric healthcare to life-course healthcare strengthens female workforce participation and social equity?

India's public health architecture has historically prioritized maternal and child interventions, effectively reducing women's health to their reproductive utility. Transitioning toward a comprehensive life-course healthcare paradigm is vital to ensuring fundamental social equity, preserving economic dignity, and protecting female labor productivity across all life stages.

Unaddressed mid-life health conditions, including severe perimenopausal distress, chronic hormonal imbalances, and non-communicable diseases, disproportionately push experienced female workers into premature retirement or informal low-wage labor. When public health centers lack screening and medical interventions for mid-life issues, older women absorb severe physical debility and mental fatigue while bearing disproportionate domestic care burdens. Institutionalizing healthcare beyond childbirth—via subsidized diagnostics, workplace wellness policies, and tailored geriatric interventions—mitigates avoidable absenteeism and sustains economic security. Moreover, a life-course approach aligns with constitutional guarantees under **Article 21**, rectifying institutional gender biases that treat women's well-being as secondary once their reproductive phase concludes. Ensuring universal access to mid-life and **geriatric care** structurally empowers women to participate freely and equitably across productive economic spheres.

Evolving from reproductive-focused interventions to holistic, lifelong universal healthcare dismantles systemic vulnerabilities, elevates female agency, and ensures that women age with economic independence, personal dignity, and complete social equality.

DAILY MCQs

- With reference to menopause and perimenopause, consider the following statements:
1. Menopause is clinically diagnosed after six consecutive months of amenorrhea.
2. It occurs primarily due to the decreased ovarian production of estrogen and progesterone.
Which of the statements given above is/are correct?
A. 1 only
B. 2 only
C. Both 1 and 2
D. Neither 1 nor 2
- Consider the following statements
1. regarding mid-life health initiatives in Indian states:
2. Karnataka launched '**Ruthu Thare**', a dedicated women's health policy addressing perimenopause and menopause.
3. Tamil Nadu's perimenopause policy integrates screening and counseling services directly through Primary Health Centres (PHCs).
Which of the statements given above is/are correct?
A. 1 only
B. 2 only
C. Both 1 and 2
D. Neither 1 nor 2
- Which of the following hormones experience a natural decline during perimenopause leading to the cessation of menstrual cycles?
A. Prolactin and Oxytocin
B. Estrogen and Progesterone
C. Luteinizing Hormone and Cortisol
D. Thyroxine and Calcitonin

Answer Key • 1-B • 2-C • 3-B